

Lecture 4: Level-2 Model for inter-individual differences

February 13th

From Last Lecture...

- What are the key takeaways from descriptive analyses
 - Sample means of the estimated intercepts and slopes
 - Sample variances (i.e. standard deviations) of the estimated intercepts and slopes
 - Correlation between the estimated intercepts and slopes (i.e. covariance)?
- Time invariant predictors and descriptive analyses
- Why do we use multi-level models of change?
 - The parameters in our basic regression equation represent population estimates of interests
 - Extension of linear regression to 2-level model
- Level-1 individual model of change (individual growth model)
 - This model represents the change we expect to see in each individual during the study period (period of observation)
 - Introduction of the Level-1 model

Learning Objectives

- Level-2 model (interindividual change in trajectories)
 - Define level-2 (interindividual change) model
 - Structural component
 - Stochastic component
- Estimating fixed effects
 - What are fixed effects?
 - Interpreting fixed effects
- Composite model (level-1 and level-2): **Mixed-Effects Model**
 - Structural component
 - Stochastic component
- Discuss assignment #2
- Administrative stuff
- Class Exercises 1 and 2

Overview of Level-1 & Level-2 Models

Longitudinal Data: observe $(\mathbf{i}, Y_{ij}, \mathbf{t}_{ij}, x_{ij1}, x_{ij2}, \dots, x_{ijp})$
for subject $i = 1, \dots, m$ and $j = 1, \dots, n_i$ (**Single-Level of Grouping**)

Models for Longitudinal Data:

1. **Population Model:** Can we build **ONE model** by ignoring repeated measurements of each subjects. I.e., $Y_{ij} = h_{\theta}(t_{ij}, x_{ij1}, \dots, x_{ijp}) + \epsilon_{ij}$
[Use **LOESS or other local smoothers** to brainstorm some rough parametric models]
2. Can we compute **summary statistics [e.g., each component of θ_i]** for each subject, and then apply linear regression to these summary statistics?
3. **Individual Models:** Can we build **m individual** models, with different **θ_i** for different subjects? I.e., $Y_{ij} = h_{\theta_i}(t_{ij}, x_{ij1}, \dots, x_{ijp}) + \epsilon_{ij}$?
[Level-1 model]
→ Relation of **θ_i** [Level-2 model]

Linear Mixed-Effect Model

Longitudinal Data: observe $(Y_{ij}, x_{ij1}, x_{ij2}, \dots, x_{ijp})$
for subject $i = 1, \dots, m$ and $j = 1, \dots, n_i$

- **Population Model**:

$$Y_{ij} = \beta_0 + \beta_1 x_{ij1} + \dots + \beta_p x_{ijp} + \epsilon_{ij}$$

where the ϵ_{ij} 's are iid $N(0, \sigma^2)$.

- **Individual Models (Level-1 Model)**

$$Y_{ij} = \beta_{i0} + \beta_{i1} x_{ij1} + \dots + \beta_{ip} x_{ijp} + \epsilon_{ij}$$

where the ϵ_{ij} 's are iid $N(0, \sigma^2)$.

- **Mixed-Effects Models**: for the components of subject-specific parameters $(\beta_{i0}, \beta_{i1}, \dots, \beta_{ip})$

- **Fixed-effects**: Some of them might be constant over different i ;
- **Random-effects**: Some of them may be modeled as **independent deviation** from the population average value $(\beta_0, \beta_1, \dots, \beta_p)$.

Linear Mixed-Effect Model with p=1: Simple Linear Regression

Individual (Level-1) Models: observe (Y_{ij}, x_{ij}) for subject $i = 1, \dots, m$ and $j = 1, \dots, n_i$ with individual models, $Y_{ij} = \beta_{i0} + \beta_{i1}x_{ij} + \epsilon_{ij}$ where the ϵ_{ij} 's are iid $N(0, \sigma^2)$. Here x_{ij} might or might not be time t_{ij} .

Possible distinct Level-2 Models for (β_{i0}, β_{i1}) : Can both be constant?

→ Composite model (level-1 and level-2): Mixed-Effects Model

1. $\beta_{i0} = \beta_0 + b_{0i}$ and $\beta_{i1} \equiv \beta_1$, where b_{0i} are iid $N(0, \sigma_{b0}^2)$,
→ $Y_{ij} = (\beta_0 + b_{0i}) + \beta_1 x_{ij} + \epsilon_{ij} = \beta_0 + \beta_1 x_{ij} + b_{0i} + \epsilon_{ij}$
2. $\beta_{i0} \equiv \beta_0$ and $\beta_{i1} = \beta_1 + b_{1i}$, where b_{1i} are iid $N(0, \sigma_{b1}^2)$.
→ $Y_{ij} = \beta_0 + (\beta_1 + b_{1i})x_{ij} + \epsilon_{ij} = \beta_0 + \beta_1 x_{ij} + x_{ij}b_{1i} + \epsilon_{ij}$
3. $\beta_{i0} = \beta_0 + b_{0i}$ and $\beta_{i1} \equiv \beta_1 + b_{1i}$
 $Y_{ij} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})x_{ij} + \epsilon_{ij} = \beta_0 + \beta_1 x_{ij} + b_{0i} + x_{ij}b_{1i} + \epsilon_{ij}$
→ How to model the joint distribution of (b_{0i}, b_{1i}) ?
(3a) uncorrelated and (3b) correlated

Bivariate Normal Distribution

When $Y_{ij} = \beta_0 + \beta_1 x_{ij} + b_{0i} + x_{ij} b_{1i} + \epsilon_{ij}$, how to model the joint distribution of (b_{0i}, b_{1i}) ?

Answer: **Bivariate Normal distribution** with mean (0,0) and

covariate matrix $\Omega = \begin{pmatrix} \sigma_{b0}^2 & \rho\sigma_{b0}\sigma_{b1} \\ \rho\sigma_{b0}\sigma_{b1} & \sigma_{b1}^2 \end{pmatrix}$,

or equivalently, the joint pdf of (b_{0i}, b_{1i}) is given by

$$\frac{1}{2\pi\sigma_{b0}\sigma_{b1}\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\left(\frac{b_{0i}}{\sigma_{b0}} \right)^2 - 2\rho \frac{b_{0i}}{\sigma_{b0}} \frac{b_{1i}}{\sigma_{b1}} + \left(\frac{b_{1i}}{\sigma_{b1}} \right)^2 \right] \right\}$$

Two different scenarios:

- a) **Uncorrelated:** Assume $\rho = 0$, and only estimate $(\sigma_{b0}^2, \sigma_{b1}^2)$
- b) **Correlated:** ρ arbitrary, and we need to estimate $(\sigma_{b0}^2, \sigma_{b1}^2, \rho)$

Summary for LME Model with $p=1$

Level-1 Mode: observe (Y_{ij}, x_{ij}) for subject $i = 1, \dots, m$ and $j = 1, \dots, n_i$ with individual models, $Y_{ij} = \beta_{i0} + \beta_{i1}x_{ij} + \epsilon_{ij}$, where the ϵ_{ij} 's are iid $N(0, \sigma^2)$.

4 Possible distinct Level-2 Models:

1. $\beta_{i0} = \beta_0 + b_{0i}$ and $\beta_{i1} \equiv \beta_1$, where b_{0i} are iid $N(0, \sigma_{b0}^2)$,
2. $\beta_{i0} \equiv \beta_0$ and $\beta_{i1} = \beta_1 + b_{1i}$, where b_{1i} are iid $N(0, \sigma_{b1}^2)$.
3. $\beta_{i0} = \beta_0 + b_{0i}$ and $\beta_{i1} \equiv \beta_1 + b_{1i}$, where (b_{0i}, b_{1i}) are Bivariate Normal with mean $(0,0)$ & covariate $\Omega = \begin{pmatrix} \sigma_{b0}^2 & \mathbf{0} \\ \mathbf{0} & \sigma_{b1}^2 \end{pmatrix}$
4. $\beta_{i0} = \beta_0 + b_{0i}$ and $\beta_{i1} \equiv \beta_1 + b_{1i}$, where (b_{0i}, b_{1i}) are Bivariate Normal with mean $(0,0)$ & covariate $\Omega = \begin{pmatrix} \sigma_{b0}^2 & \rho\sigma_{b0}\sigma_{b1} \\ \rho\sigma_{b0}\sigma_{b1} & \sigma_{b1}^2 \end{pmatrix}$

Summary for LME Model with p=1 composite model

Level-1 + Level-2 = composite model = fixed-effects + random effects

- 4 distinct composite models: observe (Y_{ij}, x_{ij}) for subject $i = 1, \dots, m$ and $j = 1, \dots, n_i$. Below ϵ_{ij} are independent of both b_{0i} and b_{1i} .
 1. $Y_{ij} = \beta_0 + \beta_1 x_{ij} + b_{0i} + \epsilon_{ij}$, where b_{0i} are iid $N(0, \sigma_{b0}^2)$
 2. $Y_{ij} = \beta_0 + \beta_1 x_{ij} + x_{ij} b_{1i} + \epsilon_{ij}$, where b_{1i} are iid $N(0, \sigma_{b1}^2)$
 3. $Y_{ij} = \beta_0 + \beta_1 x_{ij} + b_{0i} + x_{ij} b_{1i} + \epsilon_{ij}$, where b_{0i} are iid $N(0, \sigma_{b0}^2)$, **uncorrelated** of b_{1i} which are iid $N(0, \sigma_{b1}^2)$ [i.e., $\rho = 0$ under #4 notation].
 4. $Y_{ij} = \beta_0 + \beta_1 x_{ij} + b_{0i} + x_{ij} b_{1i} + \epsilon_{ij}$, where b_{0i} and b_{1i} are **correlated**,
Bivariate Normal with mean $(0,0)$ & covariate $\Omega = \begin{pmatrix} \sigma_{b0}^2 & \rho \sigma_{b0} \sigma_{b1} \\ \rho \sigma_{b0} \sigma_{b1} & \sigma_{b1}^2 \end{pmatrix}$.

Choice of LME Model

4 distinct LME Models: observe (Y_{ij}, x_{ij}) for $i = 1, \dots, m$ and $j = 1, \dots, n_i$.

1. $Y_{ij} = \beta_0 + \beta_1 x_{ij} + b_{0i} + \epsilon_{ij}$, where b_{0i} are iid $N(0, \sigma_{b0}^2)$
2. $Y_{ij} = \beta_0 + \beta_1 x_{ij} + x_{ij} b_{1i} + \epsilon_{ij}$, where b_{1i} are iid $N(0, \sigma_{b1}^2)$
3. $Y_{ij} = \beta_0 + \beta_1 x_{ij} + b_{0i} + x_{ij} b_{1i} + \epsilon_{ij}$, where b_{0i} are iid $N(0, \sigma_{b0}^2)$,
uncorrelated of b_{1i} which are iid $N(0, \sigma_{b1}^2)$ [i.e., $\rho = 0$ under #4 notation].
4. $Y_{ij} = \beta_0 + \beta_1 x_{ij} + b_{0i} + x_{ij} b_{1i} + \epsilon_{ij}$, where b_{0i} and b_{1i} are **correlated** and

Bivariate Normal with mean $(0,0)$ & covariate $\Omega = \begin{pmatrix} \sigma_{b0}^2 & \rho \sigma_{b0} \sigma_{b1} \\ \rho \sigma_{b0} \sigma_{b1} & \sigma_{b1}^2 \end{pmatrix}$.

Question: With the same fixed-effects and different random effects, which of these 4 LME models for should be used in a given dataset?

Solutions: There are no unique answers here.

- Empirical plots for b_{0i} and b_{1i}
- Domain knowledge
- Model selection criterion (e.g., AIC/BIC)
- Trial and Error

Useful Special Case with p=2

Data: observe $(Y_{ij}, x_{1ij}, x_{2i})$ for subject $i = 1, \dots, m$ and $j = 1, \dots, n_i$. That is, $x_{2ij} \equiv x_{2i}$ is constant over j (e.g., Female/Male).

Approach 1 (useful for practice):

- **Level-1 models:** $Y_{ij} = \beta_{i0} + \beta_{i1}x_{1ij} + \epsilon_{ij}$ for $j = 1, \dots, n_i$
- **Level-2 models:** both β_{i0} and β_{i1} might be modelled as a simple linear regression for x_{2i} for $i = 1, \dots, m$:

$$\begin{aligned}\beta_{i0} &= \beta_0 + \beta_2 x_{2i} + b_{0i} \\ \beta_{i1} &= \beta_1 + \beta_3 x_{2i} + b_{1i}\end{aligned}$$

→ This yields a composite model:

$$\begin{aligned}Y_{ij} &= (\beta_0 + \beta_2 x_{2i} + b_{0i}) + (\beta_1 + \beta_3 x_{2i} + b_{1i})x_{1ij} + \epsilon_{ij} \\ &= \beta_0 + \beta_1 x_{1ij} + \beta_2 x_{2i} + \beta_3 x_{1ij} x_{2i} + b_{0i} + b_{1i} x_{1ij} + \epsilon_{ij}\end{aligned}$$

Remarks: there are actually 4 LME models as in previous slides:

- (i) If we treat $b_{0i} \equiv 0$ or $b_{1i} \equiv 0$, we get two more LME models;
- (ii) Two more models whether b_{0i} and b_{1i} are uncorrelated or correlated

****Useful Special Case with p=2** (cont.)**

Data: observe $(Y_{ij}, x_{1ij}, x_{2i})$ for subject $i = 1, \dots, m$ and $j = 1, \dots, n_i$. That is, $x_{2ij} \equiv x_{2i}$ is constant over j .

Approach 2 (useful for theory): Define the interaction $x_{3ij} = x_{1ij}x_{2ij}$. Then model $(Y_{ij}, x_{1ij}, x_{2ij} \equiv x_{2i}, x_{3ij})$ directly:

- **Level-1 Model:** $Y_{ij} = \beta_{i0} + \beta_{i1}x_{1ij} + \beta_{i2}x_{2ij} + \beta_{i3}x_{3ij} + \epsilon_{ij}$
- **Level-2 model:** $\beta_{i0} = \beta_0 + b_{0i}, \beta_{i1} = \beta_1 + b_{1i}, \beta_{i2} \equiv \beta_2, \beta_{i3} \equiv \beta_3$

→ This yields a composite model:

$$\begin{aligned} Y_{ij} &= (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})x_{1ij} + \beta_2x_{2i} + \beta_3x_{1ij}x_{2i} + \epsilon_{ij} \\ &= \beta_0 + \beta_1x_{1ij} + \beta_2x_{2i} + \beta_3x_{1ij}x_{2i} + b_{0i} + b_{1i}x_{1ij} + \epsilon_{ij} \end{aligned}$$

This is equivalent to Approach #1.

Another useful special case

Longitudinal Data: observe $(Y_{ij}, \mathbf{x}_{1i}, \dots, \mathbf{x}_{pi})$ for $i = 1, \dots, m$ and $j = 1, \dots, n_i$. That is, $\mathbf{x}_{1ij} \equiv \mathbf{x}_{1i}, \dots, \mathbf{x}_{pij} \equiv \mathbf{x}_{pi}$ for all $j = 1, \dots, n_i$.

Linear-Mixed Effect Model: two different ways to define it:

- **Level-1 Model + Level 2 Model:**

- **Level-1 Model:** $Y_{ij} = \mu_i + \epsilon_{ij}$ for $i = 1, \dots, m$ and $j = 1, \dots, n_i$

- **Level-2 model:**

$$\mu_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi} + \mathbf{b}_i \text{ for } i = 1, \dots, m$$

- **Composite model (Fixed-effects+ random effects):**

$$Y_{ij} = \beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi} + \mathbf{b}_i + \epsilon_{ij},$$

where $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^T$ are fixed effects, the random effects b_i are iid with $N(0, \sigma_b^2)$, independent of ϵ_{ij} which are iid with $N(0, \sigma^2)$

Notation in Theories and Methods

- **Longitudinal Data** (Single-level Of Grouping):
observe $(Y_{ij}, x_{ij1}, x_{ij2}, \dots, x_{ijp})$ for subject $i = 1, \dots, m$
and $j = 1, \dots, n_i$
- **Linear-Mixed Effect Model** (composite model)
$$Y_{ij} = \beta_1 x_{ij1} + \dots + \beta_p x_{ijp} + b_{1i} z_{ij1} + \dots + b_{qi} z_{ijq} + \epsilon_{ij},$$

where
 - $\{z_{ij1}, \dots, z_{ijq}\}$ is a subset of $\{x_{ij1}, \dots, x_{ijp}\}$
 - The p-dimensional vector $\beta = (\beta_1, \dots, \beta_p)^T$ are fixed effects,
 - The q-dim random effects vectors $b_i = (b_{1i}, \dots, b_{qi})^T$ are iid with **Multivariate Normal with mean 0 and covariance Ω** , for $i = 1, \dots, m$
 - The residuals/noises ϵ_{ij} are iid with $N(0, \sigma^2)$

Additive Noise Model

Linear-Mixed Effect Model (composite model)

$$Y_{ij} = \beta_1 x_{ij1} + \cdots + \beta_p x_{ijp} + \textcolor{red}{b_{1i} z_{ij1}} + \cdots + \textcolor{red}{b_{qi} z_{ijq}} + \epsilon_{ij},$$

- **Key statistical Idea:**

Observed Data = True Value + Noise

where

Noise = Subject-Random-Effects + Measurement Error

Intercepts or not?

- Intercepts in a model is not necessarily mandatory
- In some real-world applications, the model might not include intercepts.
- From the theoretical viewpoint, we can create a new independent variable $x_{ij1} \equiv 1$, whose coefficient will be intercept.

Example: Ball weight

- Three NYU students, “Alice”, “Bob” and “Charlie”, were asked to work together to find the weights of two balls, A and B, using a scale with random measurement errors.
- When “Alice” measured one ball at a time, the weights of A and B were 1 lbs and 4 lbs. However, if Alice measured both balls simultaneously, then the total weight ($A+B$) was 4 lbs.
- When “Bob” repeated the above process, the observed weights for A, B and ($A+B$) were 2,2,3 lbs.
- When “Charlie” repeated the above process, the observed weights for A, B and ($A+B$) were 3,3,5 lbs.

Questions: find the best estimates of the weights of balls A and B.

Data

	A	B	A+B
Alice	1	4	4
Bob	2	2	3
Charlie	3	3	5

- **Model: Observed Data = True Value + Noise** where
Noise = Subject-Random-Effects + Measurement Error
- Can we write down these math equations?

Data → Model

	A	B	A+B
Alice	1	4	4
Bob	2	2	3
Charlie	3	3	5

- **Model: Observed Data = True Value + Noise** where
Noise = Subject-Random-Effects + Measurement Error

$$1 = A + b_1 + \epsilon_{11}$$

$$4 = B + b_1 + \epsilon_{12}$$

$$4 = A + B + b_1 + \epsilon_{13}$$

$$2 = A + b_2 + \epsilon_{21}$$

$$2 = B + b_2 + \epsilon_{22}$$

$$3 = A + B + b_2 + \epsilon_{23}$$

$$3 = A + b_3 + \epsilon_{31}$$

$$3 = B + b_3 + \epsilon_{32}$$

$$5 = A + B + b_3 + \epsilon_{33}$$

Data → Model → Long format

ID	Y	X1	X2
Alice	1	1	0
Alice	4	0	1
Alice	4	1	1
Bob	2	1	0
Bob	2	0	1
Bob	3	1	1
Charlie	3	1	0
Charlie	3	0	1
Charlie	5	1	1

R code & output

```
> data0 <- read.csv("ball.csv", head=T, sep=",")
> summary(lme01 <- lmer(Y ~ X1 + X2-1 + (1 | ID1), data =
data0))
```

Linear mixed model fit by REML ['lmerMod']

Formula: Y ~ X1 + X2 - 1 + (1 | ID1)

Data: data0

REML criterion at convergence: 23

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.15807	-0.47934	0.07448	0.60424	1.42294

Random effects:

Groups	Name	Variance	Std.Dev.
ID1	(Intercept)	0.1644	0.4054
	Residual	0.8517	0.9229

Number of obs: 9, groups: ID1, 3

Fixed effects:

	Estimate	Std. Error	t value
X1	1.6264	0.4606	3.531
X2	2.6264	0.4606	5.702

Correlation of Fixed Effects:

X1	
X2	-0.338

For Alice/Bob/Charlie:

$$\hat{A} = 1.6264$$

$$\hat{B} = 2.6264$$

Example: Ball weight (cont.)

- Suppose three other students, “David”, “Emily” and “Frank”, were also asked to work together to find the weights of two balls, A and B, using a scale with random measurement errors. However, their team strategies are different.
- “David ” measured the weight of ball “A” three times, and the observed weights are 1,2,3 lbs.
- “Emily” measured the weight of ball “B” three times, and the observed weights are 4,2,3 lbs.
- “Frank” measured the combined weight of “(A+B)” three times, and the observed weights are 4,3,5 lbs.

Questions: find the best estimates of the weights of balls A and B.

Data

	David	Emily	Frank
	A	B	A+B
Alice	1	4	4
Bob	2	2	3
Charlie	3	3	5

- **Model: Observed Data = True Value + Noise** where
Noise = Subject-Random-Effects + Measurement Error
- Can we write down these math equations for
“David/Emily/Frank”?

Data → Model

	David	Emily	Frank
	A	B	A+B
Alice	1	4	4
Bob	2	2	3
Charlie	3	3	5

- Model for Alice/Bob/Charlie:

$$1 = A + b_1 + \epsilon_{11}$$

$$4 = B + b_1 + \epsilon_{12}$$

$$4 = A + B + b_1 + \epsilon_{13}$$

$$2 = A + b_2 + \epsilon_{21}$$

$$2 = B + b_2 + \epsilon_{22}$$

$$3 = A + B + b_2 + \epsilon_{23}$$

$$3 = A + b_3 + \epsilon_{31}$$

$$3 = B + b_3 + \epsilon_{32}$$

$$5 = A + B + b_3 + \epsilon_{33}$$

- Model for David/Emily/Frank

$$1 = A + b_1 + \epsilon_{11}$$

$$2 = A + b_1 + \epsilon_{12}$$

$$3 = A + b_1 + \epsilon_{13}$$

$$4 = B + b_2 + \epsilon_{21}$$

$$2 = B + b_2 + \epsilon_{22}$$

$$3 = B + b_2 + \epsilon_{23}$$

$$4 = A + B + b_3 + \epsilon_{31}$$

$$3 = A + B + b_3 + \epsilon_{32}$$

$$5 = A + B + b_3 + \epsilon_{33}$$

Long format (equivalent)

(Sort by ID1) (Sort by ID2)

ID1	ID2	Y	X1	X2		ID1	ID2	Y	X1	X2
Alice	David	1	1	0	↔	Alice	David	1	1	0
Alice	Emily	4	0	1	↗	Bob	David	2	1	0
Alice	Frank	4	1	1	↘	Charlie	David	3	1	0
Bob	David	2	1	0	↖	Alice	Emily	4	0	1
Bob	Emily	2	0	1	↔	Bob	Emily	2	0	1
Bob	Frank	3	1	1	↘	Charlie	Emily	3	0	1
Charlie	David	3	1	0	↖	Alice	Frank	4	1	1
Charlie	Emily	3	0	1	↗	Bob	Frank	3	1	1
Charlie	Frank	5	1	1	↔	Charlie	Frank	5	1	1

R code & output

```
> summary(lme01 <- lmer(Y ~ X1 + X2-1 + (1 | ID2), data = data0))
```

Linear mixed model fit by REML ["lmerMod"]

Formula: Y ~ X1 + X2 - 1 + (1 | ID2)

Data: data0

REML criterion at convergence: 23.2

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.3333	-0.6667	0.3333	0.6667	1.3333

Random effects:

Groups	Name	Variance	Std.Dev.
ID2	(Intercept)	1.745e-08	0.0001321
	Residual	1.000e+00	1.0000000

Number of obs: 9, groups: ID2, 3

Fixed effects:

	Estimate	Std. Error	t value
X1	1.6667	0.4714	3.536
X2	2.6667	0.4714	5.657

Correlation of Fixed Effects:

X1	
X2	-0.500

optimizer (nloptwrap) convergence code: 0 (OK)

Hessian is numerically singular: parameters are not uniquely determined

Warning message:

In checkConv(attr(opt, "derivs"), opt\$par, ctrl = control\$checkConv, :

Hessian is numerically singular: parameters are not uniquely determined

For Alice/Bob/Charlie:

$$\hat{A} = 1.6264$$

$$\hat{B} = 2.6264$$

For David/Emily/Frank:

$$\hat{A} = 1.6667$$

$$\hat{B} = 2.6667$$

How to select random-effects in LME

There are many possible (linear or nonlinear) mixed-effects models for a given dataset. Which one should be used?

- Empirical plots, e.g., 95% Confidence Intervals for individual model parameters.
- Domain knowledge
- Model selection criterion (e.g., AIC/BIC)
- Trial and Error: if the most complicated model does not converge or does not make sense, then try a simpler model.

Linear Mixed-Effect Model

Longitudinal Data: observe $(\mathbf{i}, Y_{ij}, \mathbf{t}_{ij}, x_{ij1}, x_{ij2}, \dots, x_{ijp})$
for subject $i = 1, \dots, m$ and $j = 1, \dots, n_i$

- **Population Model:**

$$Y_{ij} = \beta_0 + \beta_1 x_{ij1} + \dots + \beta_p x_{ijp} + \epsilon_{ij}$$

where the ϵ_{ij} 's are iid $N(0, \sigma^2)$.

- **Level-2 model on $\theta_i = (\theta_{i1}, \theta_{i2}, \dots, \theta_{ip})$:**

Check 95% confidence intervals (CIs) for each component of θ_i over different individuals:

- **When those 95% CIs overlaps**, such components of θ_i might be treated as the same (**fixed effects**)
- **When those 95% CIs are very different**, such components of θ_i might be treated as iid random variables (**random effects**)

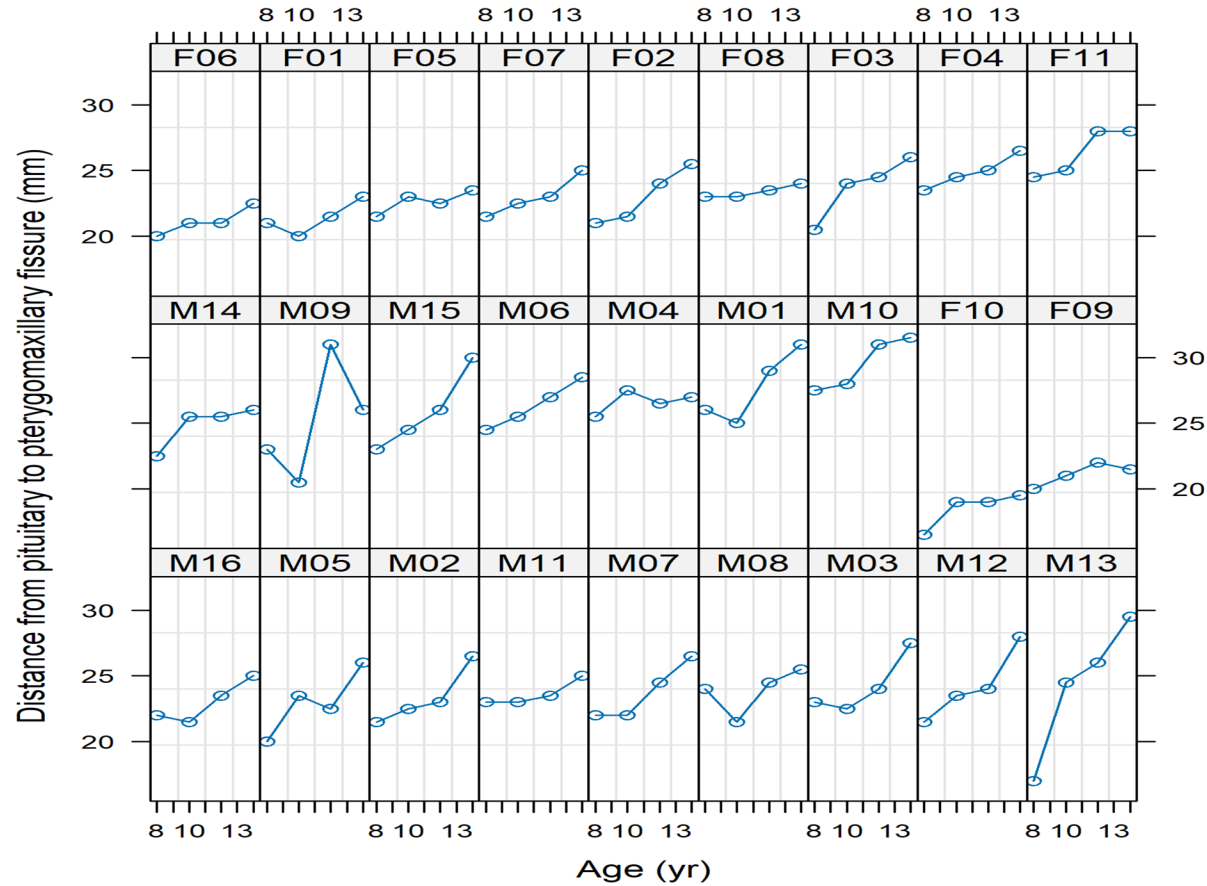
Review - Lecture 1

- Potthoff and Roy (1964) gave a set of measurements of the distance from the pituitary gland to the pterygo-maxillary fissure taken every two years from 8 years of age until 14 years of age on a sample of 27 children --- 16 males and 11 females
- The data were collected by orthodontists from x-rays of the children's skulls. The pituitary gland and the pterygomaxillary fissure are two easily located points on these x-rays.

Scientific Questions:

- **Are the growth patterns for males/females different?**

Data



R code

```
### You need to install the R package "nlme" first  
library(nlme);
```

```
### data
```

```
> head(Orthodont)
```

```
Grouped Data: distance ~ age | Subject
```

	distance	age	Subject	Sex
1	26.0	8	M01	Male
2	25.0	10	M01	Male
3	29.0	12	M01	Male
4	31.0	14	M01	Male
5	21.5	8	M02	Male
6	22.5	10	M02	Male

```
# plot them
```

```
plot(Orthodont)
```

4 LME Model and R codes: #1

Model #1: (only intercept is random effects)

$$Y_{ij} = \beta_0 + \beta_1(\text{Age}_{ij} - 11) + \beta_2 * \text{Female} + \beta_3 * \text{Female} * (\text{Age}_{ij} - 11) \\ + b_{0i} + \epsilon_{ij}$$

where b_{0i} are iid $N(0, \sigma_{b0}^2)$

R codes:

- nlme

```
lme.A01 <- lme(distance ~ Sex*I(age-11), data =Orthodont,  
random=~1);
```

- Lme4

```
lme.B01 <- lmer(distance ~ Sex*I(age-11) + ( 1 |Subject), data =  
Orthodont)
```

4 LME Model and R codes: #2

Model #2: (only slope is random effects)

$$Y_{ij} = \beta_0 + \beta_1(\text{Age}_{ij} - 11) + \beta_2 * \text{Female} + \beta_3 * \text{Female} * (\text{Age}_{ij} - 11) \\ + b_{1i}(\text{Age}_{ij} - 11) + \epsilon_{ij}$$

where b_{1i} are iid $N(0, \sigma_{b1}^2)$

R codes:

- nlme

```
lme.A02 <- lme(distance ~ Sex*I(age-11), data =Orthodont,  
random=~I(age-11)-1);
```

- lme4

```
lme.B02 <- lmer(distance ~ Sex*I(age-11) + ( 0+I(age-11)  
|Subject), data = Orthodont)
```

4 LME Model and R codes: #3

Model #3: (both intercepts and slopes are random effects, but uncorrelated)

$$Y_{ij} = \beta_0 + \beta_1(\text{Age}_{ij} - 11) + \beta_2 * \text{Female} + \beta_3 * \text{Female} * (\text{Age}_{ij} - 11) + b_{0i} + b_{1i}(\text{Age}_{ij} - 11) + \epsilon_{ij}$$

where b_{0i} are iid $N(0, \sigma_{b0}^2)$, uncorrelated with b_{1i} iid $N(0, \sigma_{b1}^2)$

R codes:

- nlme

```
lme.A03 <- lme(distance ~ Sex*I(age-11), data = Orthodont,  
random= pdDiag(~I(age-11)));
```

- lme4

```
lme.B03 <- lmer(distance ~ Sex*I(age-11) + (1|Subject) +  
(0+ I(age-11) |Subject), data = Orthodont)
```

4 LME Model and R codes: #4

Model #4: (both intercepts and slopes are random effects, but uncorrelated)

$$Y_{ij} = \beta_0 + \beta_1(\text{Age}_{ij} - 11) + \beta_2 * \text{Female} + \beta_3 * \text{Female} * (\text{Age}_{ij} - 11) + b_{0i} + b_{1i}(\text{Age}_{ij} - 11) + \epsilon_{ij}$$

where b_{0i} and b_{1i} are **correlated** and Bivariate Normal with mean (0,0) and covariate $\Omega = \begin{pmatrix} \sigma_{b0}^2 & \rho\sigma_{b0}\sigma_{b1} \\ \rho\sigma_{b0}\sigma_{b1} & \sigma_{b1}^2 \end{pmatrix}$.

R codes:

- nlme

```
lme.A04 <- lme(distance ~ Sex*I(age-11), data =Orthodont,
random= ~I(age-11) );
```

- lme4

```
lme.B04 <- lmer(distance ~ Sex*I(age-11) + ( I(age-11) |Subject), data =
Orthodont)
```

Example from Dr. Cook's lectures
Level-2 model
(interindividual change in trajectories)

Review - Class Exercise (Lecture 3)

What is the association between age and alcohol use? Does this association vary by age?

Write and describe the level one equation

$$ALCUSE_{ij} = \pi_{0i} + \pi_{1i}(Age_{ij} - 14) + \varepsilon_{it}$$

- Y_{ij} : The value of alcohol use for adolescent i at time j
- π_{0i} : Adolescent's i 's true alcohol use at age 14
- π_{1i} : The slope (annual rate of change for adolescent i)
- ε_{ij} : Random measurement error for adolescent i at time j

Level-2 individual model of change

(Singer and Willet pages 57-58)

- This model allows us to examine the association between individual growth parameters and time-invariant covariates
 - E.g. how does type of program (e.g. control of treatment) influence initial cognitive scores and the rate of change in cognitive scores across time?
- Like level-1, we are attempting to describe a population process utilizing sample data

Level-2 individual model of change

(Singer and Willet pages 57-58)

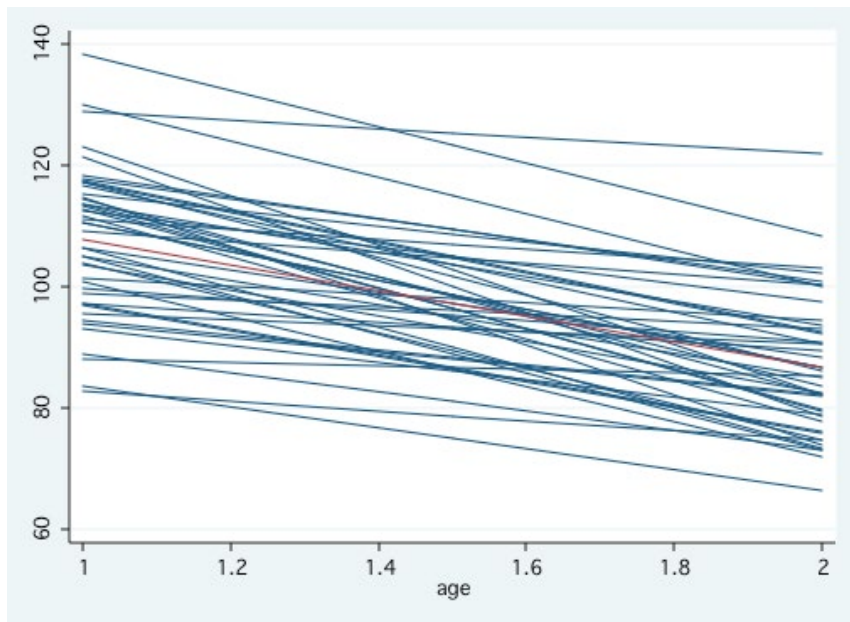
4 important features

- **(1&2)** We are modeling the population distribution of the individual growth parameters outlined in the Level-1 equation. Thus, these growth parameters (intercept $[\pi_{0i}]$ and slope $[\pi_{1i}]$) become the outcomes for the level-2 equations
- **(3)** Each of the equations must specify an association between the individual growth parameters and the time invariant covariate
- **(4)** Each level-2 model must allow individuals to vary in their individual growth trajectories

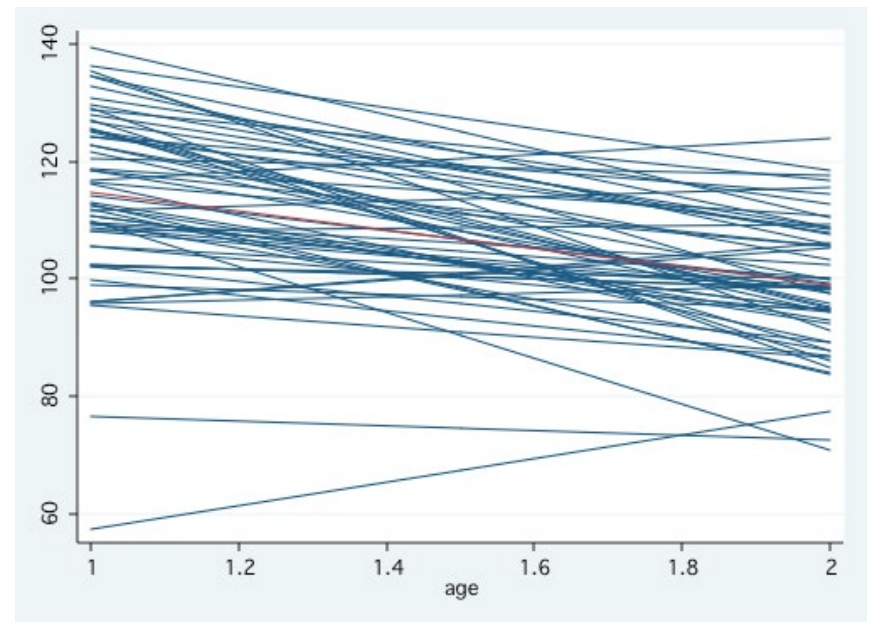
Level-2 individual model of change

(Singer and Willet page 59)

Program = 0



Program = 1



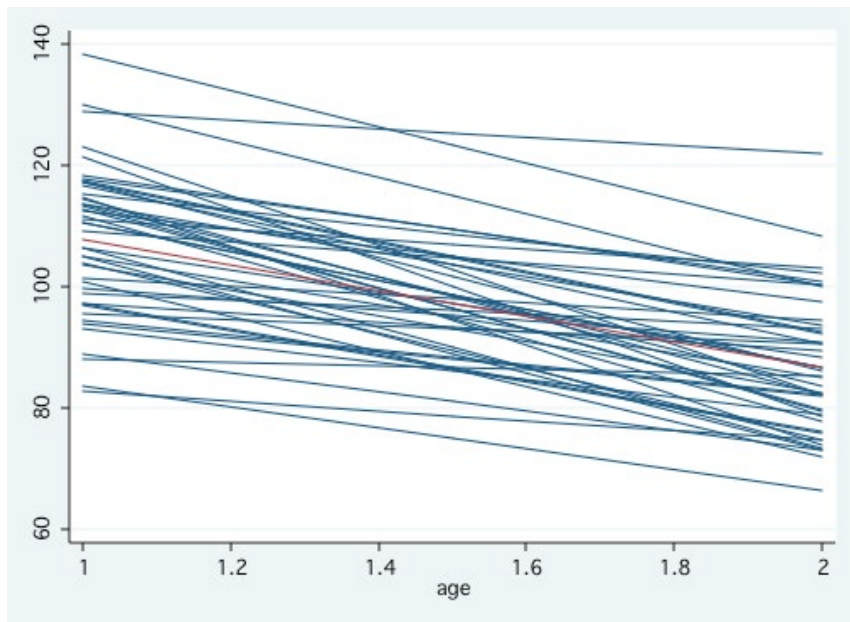
- OLS Regression by program with mean trajectory line
- Are there differences between treatment and control in terms of the intercept and slope?

(see entire code at <https://stats.idre.ucla.edu/stata/examples/alda/chapter3/applied-longitudinal-data-analysis-modeling-change-and-event-occurrenceby-judith-d-singer-and-john-b-willett-chapter-3-introducing-the-multilevel-model-for/>)

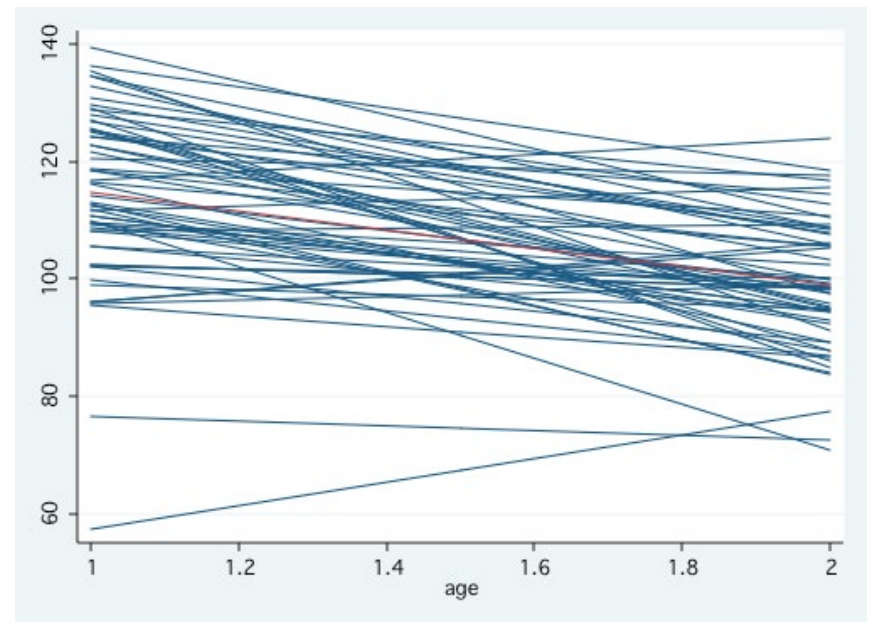
Level-2 individual model of change

(Singer and Willet page 59)

Program = 0



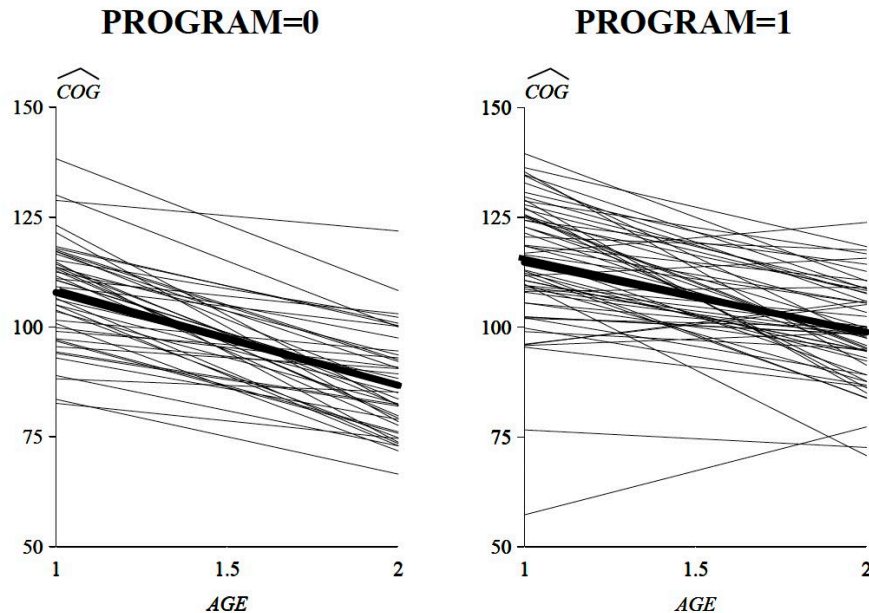
Program = 1



- The African-American infants who were given the treatment had higher initial cognitive scores and declined at a slower rate than those African-American infants in the control group

(see entire code at <https://stats.idre.ucla.edu/stata/examples/alda/chapter3/applied-longitudinal-data-analysis-modeling-change-and-event-occurrenceby-judith-d-singer-and-john-b-willett-chapter-3-introducing-the-multilevel-model-for/>)

Further developing the level-2 submodel for interindividual differences in change



Program participants tend to have:

- Higher scores at age 1 (higher initial status)
- Less steep rates of decline (shallower slopes)
- But these are only overall trends—there's great interindividual heterogeneity

Four desired features of the level-2 submodel(s)

1. Outcomes are the level-1 individual growth parameters π_{0i} and π_{1i}
2. Need two level-2 submodels, one per growth parameter (one for initial status, one for change)
3. Each level-2 submodel must specify the relationship between a level-1 growth parameter and predictor(s), here PROGRAM
 - 🎮 We need to specify a functional form for these relationships at level-2 (beginning with linear but ultimately becoming more flexible)
4. Each level-2 submodel should allow individuals with common predictor values to nevertheless have different individual change trajectories
 - 🎮 We need stochastic variation at level-2, too
 - 🎮 Each level-2 model will need its own error term, and we will need to allow for covariance across level-2 errors

Level-2 individual model of change

(Singer and Willet page 59)

$$\pi_{0i} = \gamma_{00} + \gamma_{01}PROGRAM_i + \zeta_{0i}$$

$$\pi_{1i} = \gamma_{10} + \gamma_{11}PROGRAM_i + \zeta_{1i}.$$

- γ_{00} and γ_{10} : Level-2 intercepts
- γ_{01} and γ_{11} : Level-2 slopes
- **Program_i**: Time invariant covariate (predictor)
- ϵ_{0i} and ϵ_{1i} : Residual variance that allows individual variations in the level 1 growth parameters (π_{0i} and π_{1i})

Level-2 individual model of change

(Singer and Willet page 60-61)

$$\pi_{0i} = \gamma_{00} + \gamma_{01}PROGRAM_i + \zeta_{0i}$$

$$\pi_{1i} = \gamma_{10} + \gamma_{11}PROGRAM_i + \zeta_{1i}.$$

- Let's say program = 0

$$\pi_{0i} = \gamma_{00} + \zeta_{0i} \text{ and } \pi_{1i} = \gamma_{10} + \zeta_{1i}$$

- π_{0i} (initial status) and π_{1i} (annual rate of change) are centered around level-2 growth parameters -- γ_{00} (average true initial status) and γ_{10} (true annual rate of change)
- Taken together, this model is answering the question of: **What is the average true trajectory of change in the population for children who did not receive the early intervention program?**

Level-2 individual model of change

(Singer and Willet page 60-61)

$$\pi_{0i} = \gamma_{00} + \gamma_{01}PROGRAM_i + \zeta_{0i}$$

$$\pi_{1i} = \gamma_{10} + \gamma_{11}PROGRAM_i + \zeta_{1i}$$

Hypothesized differences in the average true annual rate of change between treatment and control

- Let's say program = 1

$$\pi_{0i} = (\gamma_{00} + \gamma_{01}) + \zeta_{0i} \text{ and } \pi_{1i} = (\gamma_{10} + \gamma_{11}) + \zeta_{1i}$$

Hypothesized differences in average initial status between treatment and control

- π_{0i} (initial status) and π_{1i} (annual rate of change) are centered around level-2 growth parameters -- $\pi_{0i} = (\gamma_{00} + \gamma_{01}) + \zeta_{0i}$ and $\pi_{1i} = (\gamma_{10} + \gamma_{11}) + \zeta_{1i}$
- Taken together, this model is answering the question of: What is the difference in the average trajectory of true change associated with being in the program?

Level-2 submodels for systematic interindividual differences in change

For the level-1 intercept (initial status)

$$\pi_{0i} = \gamma_{00} + \gamma_{01} \text{PROGRAM} + \zeta_{0i}$$

For the level-1 slope (rate of change)

$$\pi_{1i} = \gamma_{10} + \gamma_{11} \text{PROGRAM} + \zeta_{1i}$$

Symbol	Definition	Illustrative interpretation
γ_{00}	Population average of the level-1 intercepts, π_{0i} , for individuals with a level-2 predictor value of 0.	Population average true initial status for nonparticipants.
γ_{01}	Population average difference in level-1 intercept, π_{0i} , for a 1-unit difference in the level-2 predictor.	Difference in population average true initial status between participants and nonparticipants.
γ_{10}	Population average of the level-1 slopes, π_{1i} , for individuals with a level-2 predictor value of 0.	Population average annual rate of true change for nonparticipants.
γ_{11}	Population average difference in level-1 slope, π_{1i} , for a 1-unit difference in the level-2 predictor.	Difference in population average annual rate of true change between participants and non-participants.

Key to remembering subscripts on the gammas (the γ 's)

- **First subscript** indicates role in level-1 model (0 for intercept; 1 for slope)
- **Second subscript** indicates role in level-2 model (0 for intercept; 1 for slope)

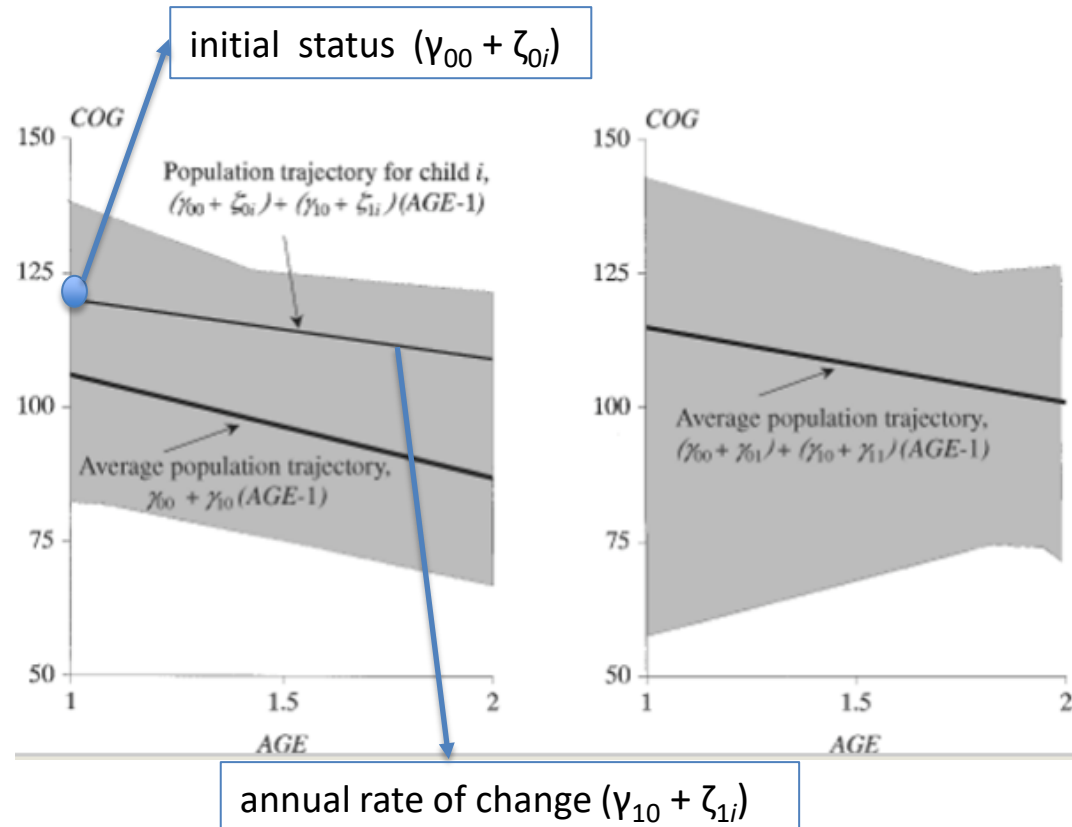
What about the zetas (the ζ 's)?

- They're level-2 residuals that permit the level-1 individual growth parameters to vary stochastically across people
- As with most residuals, we're less interested in their values than their population variances and covariances →

Stochastic part of level-2 equation

(Singer and Willet page 61-63)

- The residuals, ζ_{0i} and ζ_{1i} , make up all the the proportion of the level 2 outcomes (the intercept and slope) that remain unexplained by the level 2 predictor
- True trajectory for a typical child
 - Has a true intercept
 - Has a true declining annual rate of change
- There are many true trajectories as evidenced by the shaded area
- Level 2 is very interested in understanding the heterogeneity in growth parameters
 - Variation of the individual growth parameters around the average population trajectory
- With this part of the model we are asking, “How much heterogeneity in true change remains after controlling for the effects of the predictor (i.e. program)?”



Understanding the stochastic components of the level-2 submodels

$$\pi_{0i} = \gamma_{00} + \gamma_{01} \text{PROGRAM} + \zeta_{0i}$$

$$\pi_{1i} = \gamma_{10} + \gamma_{11} \text{PROGRAM} + \zeta_{1i}$$

General idea:

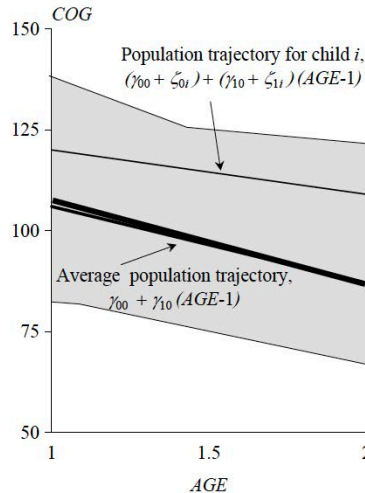
- Model posits the existence of an average population trajectory for each program group
- Because of the level-2 residuals, each child i has his own true change trajectory (defined by π_{0i} and π_{1i})
- Shading suggests the existence of many true population trajectories, one per child

Assumptions about the level-2 residuals:

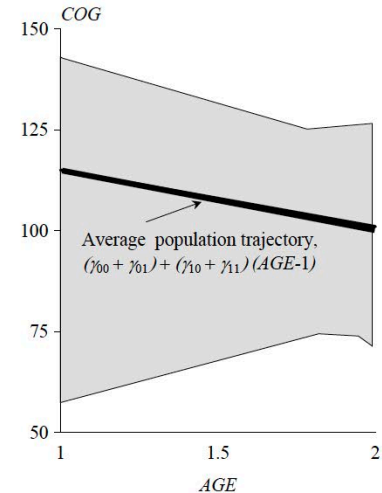
initial status $\left[\begin{matrix} \zeta_{0i} \\ \zeta_{1i} \end{matrix} \right] \sim N \left(\left[\begin{matrix} 0 \\ 0 \end{matrix} \right], \left[\begin{matrix} \sigma_0^2 & \sigma_{01} \\ \sigma_{10} & \sigma_1^2 \end{matrix} \right] \right)$

rate of change

PROGRAM=0



PROGRAM=1



Symbol	Definition	Illustrative interpretation
σ_0^2	Level-2 residual variance in true intercept, π_{0i} , across all individuals in the population.	Population residual variance of true initial status, controlling for program participation.
σ_1^2	Level-2 residual variance in true slope, π_{1i} , across all individuals in the population.	Population residual variance of true rate of change, controlling for program participation.
σ_{01}	Level-2 residual covariance between true intercept, π_{0i} , and true slope, π_{1i} , across all individuals in the population.	Population residual covariance between true initial status and true annual rate of change, controlling for program participation.

Estimating fixed effects

What are fixed effects?

(Singer and Willet pages 68-70)

- Fixed effects are responsible for assessing interindividual differences in change trajectory by values of the level-2 predictors
 - γ_{00} , γ_{01} , γ_{10} , and γ_{11}
- In our example, the fixed effects assess the association between individual growth parameters and program participation group

Estimating fixed effects?

(Singer and Willet page 69; figure 3.3)

Table 3.3: Results of fitting a multilevel model for change to the early intervention data (n=103)

		Parameter	Estimate	ase	z
Fixed Effects					
Initial status, η_{0i}	Intercept	γ_{00}	107.84***	2.04	52.97
	<i>PROGRAM</i>	γ_{01}	6.85*	2.71	2.53
Rate of change, η_{1i}	Intercept	γ_{10}	-21.13***	1.89	-11.18
	<i>PROGRAM</i>	γ_{11}	5.27*	2.52	2.09
Variance Components					
Level 1:	Within-person, ϵ_{ij}	σ_{ϵ}^2	74.24***	10.34	7.17
Level 2:	In initial status, ζ_{0i}	σ_0^2	124.64***	27.38	4.55
	In rate of change, ζ_{1i}	σ_1^2	12.29	30.50	0.40
	Covariance between ζ_{0i} and ζ_{1i}	σ_{01}	-36.41	22.74	-1.60

$\sim p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

This model predicts cognitive functioning between ages 1 and 2 years as a function of (*AGE*-1) (at level-1) and *PROGRAM* (at level-2).

Note: Full ML, HLM.

True initial status of cognitive ability at age 1 for the average non-participant; for the average participant in the program the score is 6.85 points higher.

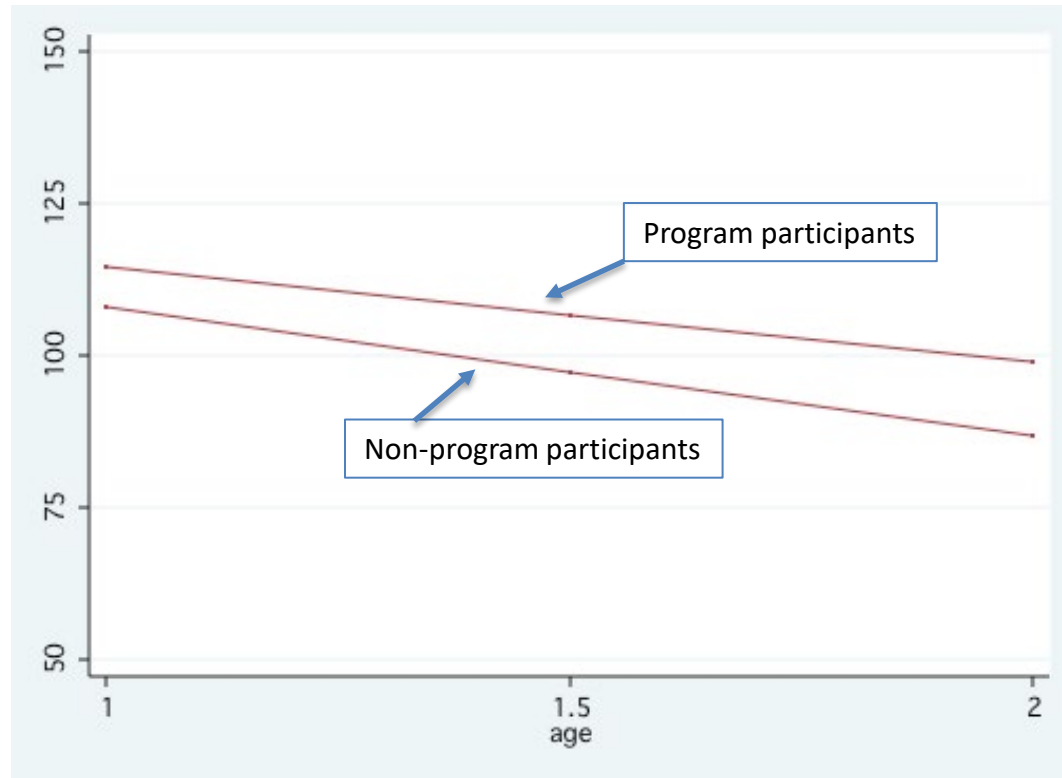
True annual rate of change in cognitive ability at age 1 for the average non-participant; for the average participant its 5.27 points higher.

Estimating fixed effects?

(Singer and Willet page 71; figure 3.5)

When $PROGRAM = 0$: $\hat{\pi}_{0i} = 107.84 + 6.85(0) = 107.84$
 $\hat{\pi}_{1i} = -21.13 + 5.27(0) = -21.13$

When $PROGRAM = 1$: $\hat{\pi}_{0i} = 107.84 + 6.85(1) = 114.69$
 $\hat{\pi}_{1i} = -21.13 + 5.27(1) = -15.86$



Examining estimated variance components

General idea:

- *Variance components quantify the amount of residual variation left—at either level-1 or level-2—that is potentially explainable by other predictors not yet in the model.*
- *Interpretation is easiest when comparing different models that each have different predictors (which we will soon do...).*

Level-1 residual variance (74.24***):

- *Summarizes within-person variability in outcomes around individuals' own trajectories (usually non-zero)*
- *Here, we conclude there is some within-person residual variability*

		Parameter	Estimate	ast	z
Variance Components					
Level 1:	Within-person, ϵ_{ij}	σ_ϵ^2	74.24***	10.34	7.17
Level 2:	In initial status, ζ_{0i}	σ_0^2	124.64***	27.38	4.55
	In rate of change, ζ_{1i}	σ_1^2	12.29	30.50	0.40
	Covariance between ζ_{0i} and ζ_{1i}	σ_{01}	-36.41	22.74	-1.60

~ $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Level-2 residual variance:

$$\begin{bmatrix} 124.64*** & -36.41 \\ -36.41 & 12.29 \end{bmatrix}$$

- *Summarizes between-person variability in change trajectories (here, initial status and growth rates) after controlling for predictor(s) (here, PROGRAM)*
- *No residual variance in rates of change to be explained (nor is there a residual covariance)*

Composite model (level-1 and
level-2)

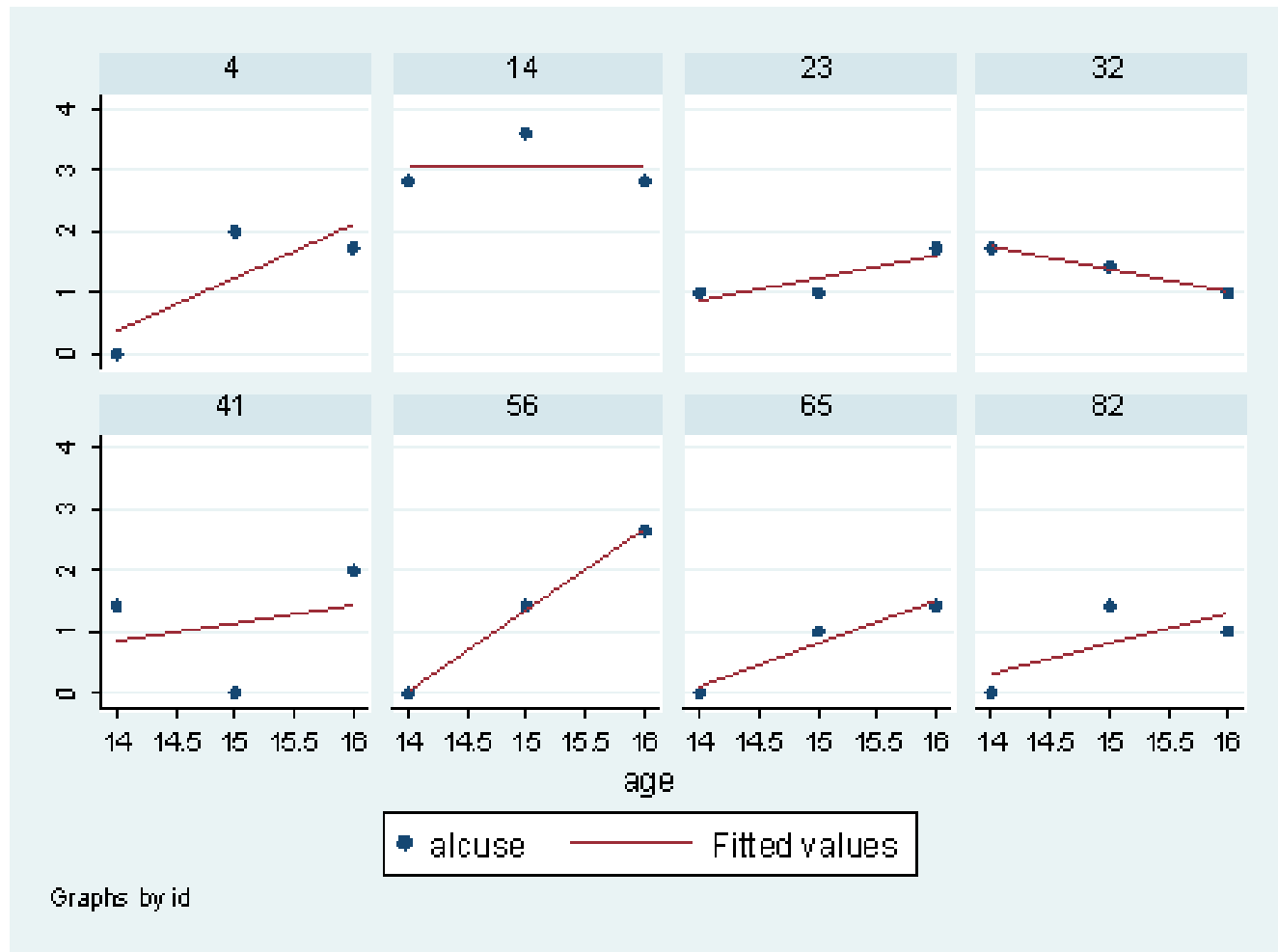
Composite model (alcohol use example)

(Singer and Willet pages 76-81)

- The outcome is alcohol use (**ALCUSE**) in the past year on an 8 point scale with higher numbers indicating more use
- The time invariant predictor is dichotomous – child of an alcoholic (**COA**) or not
- The time invariant continuous predictor is proportion of peers who use alcohol at baseline (**PEER**)
- **Age** is the time variable (13-17 years of age)

Composite model (alcohol use example)

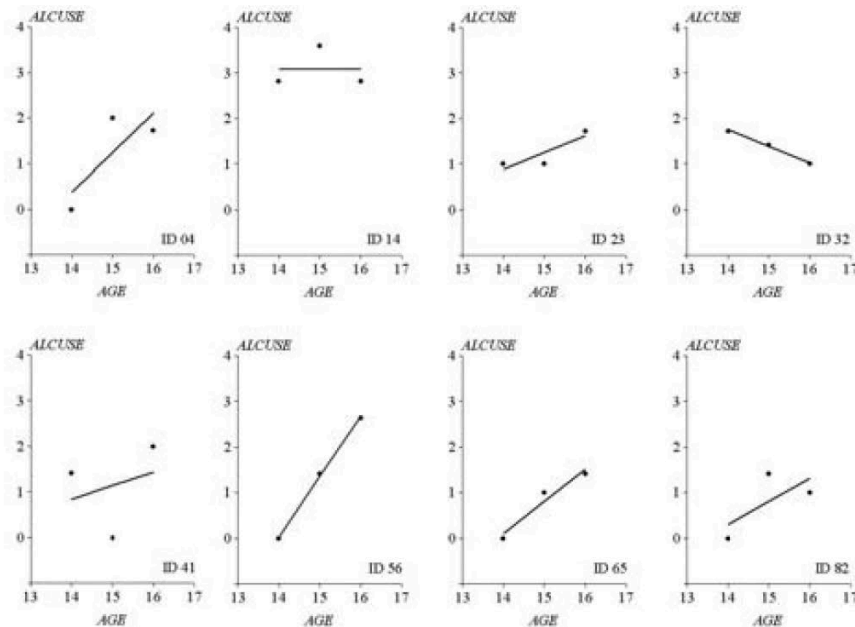
(Singer and Willett pages 76-81)



What's an appropriate functional form for the level-1 submodel? (Examining empirical growth plots with superimposed OLS trajectories)

3 features of these plots:

1. Often approximately linear (but not always increasing over time)
2. Some OLS trajectories fit well (23, 32, 56, 65)
3. Other OLS trajectories show more scatter (04, 14, 41, 82)



A linear model makes sense...

$$ALCUSE_{ij} = \pi_{0i} + \pi_{1i}(AGE_{ij} - 14) + \varepsilon_{ij} \quad \text{where } \varepsilon_{ij} \sim N(0, \sigma_{\varepsilon}^2)$$

$$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}$$

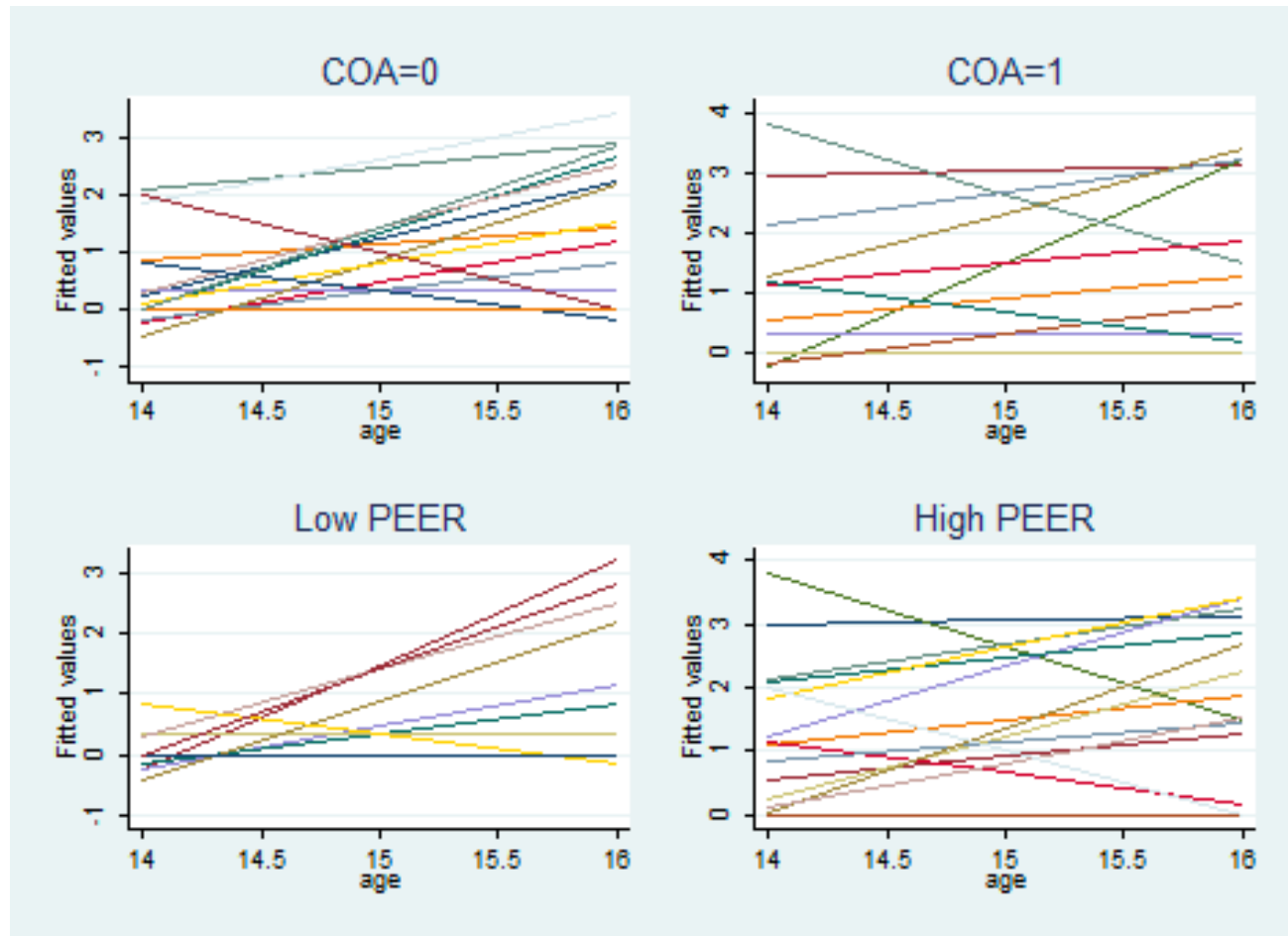
π_{0i} : i 's true initial status
(ie, when TIME=0)

π_{1i} : i 's true rate of change
per unit of TIME

ε_{ij} : portion of i 's outcome
unexplained on occasion j

Composite model (alcohol use example)

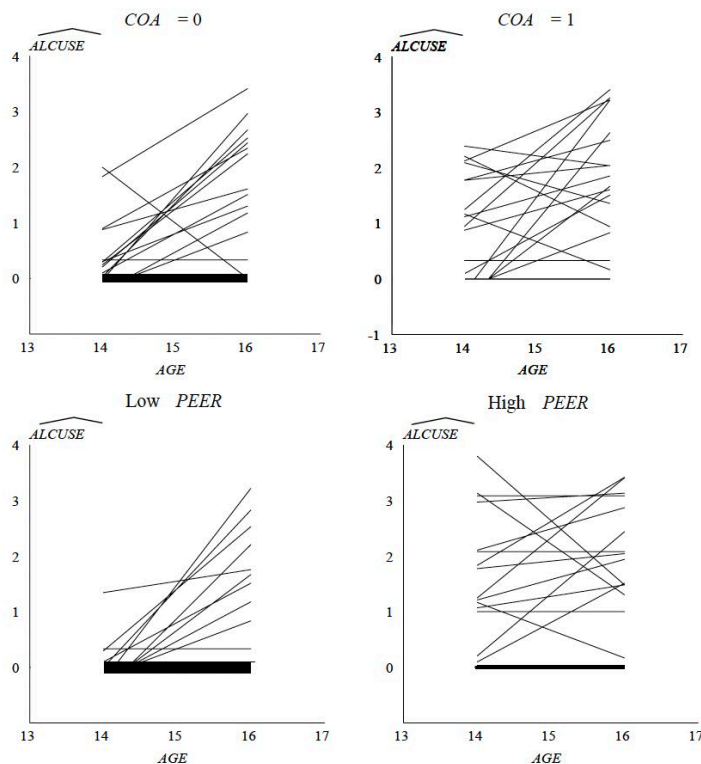
(Singer and Willet pages 76-81)



Specifying the level-2 submodels for individual differences in change

Examining variation in OLS-fitted level-1 trajectories by:

- COA:** COAs have higher intercepts but no steeper slopes
- PEER (split at mean):** Teens whose friends at age 14 drink more have higher intercepts but shallower slopes



Level-2 intercepts
Population average initial status and rate of change for a non-COA

Level-2 slopes
Effect of COA on initial status and rate of change

$$\pi_{0i} = \gamma_{00} + \gamma_{01} COA_i + \zeta_{0i} \quad (\text{for initial status})$$

$$\pi_{1i} = \gamma_{10} + \gamma_{11} COA_i + \zeta_{1i} \quad (\text{for rate of change})$$

Level-2 residuals
Deviations of individual change trajectories around predicted averages

$$\begin{bmatrix} \zeta_{0i} \\ \zeta_{1i} \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_0^2 & \sigma_{01} \\ \sigma_{10} & \sigma_1^2 \end{bmatrix} \right)$$

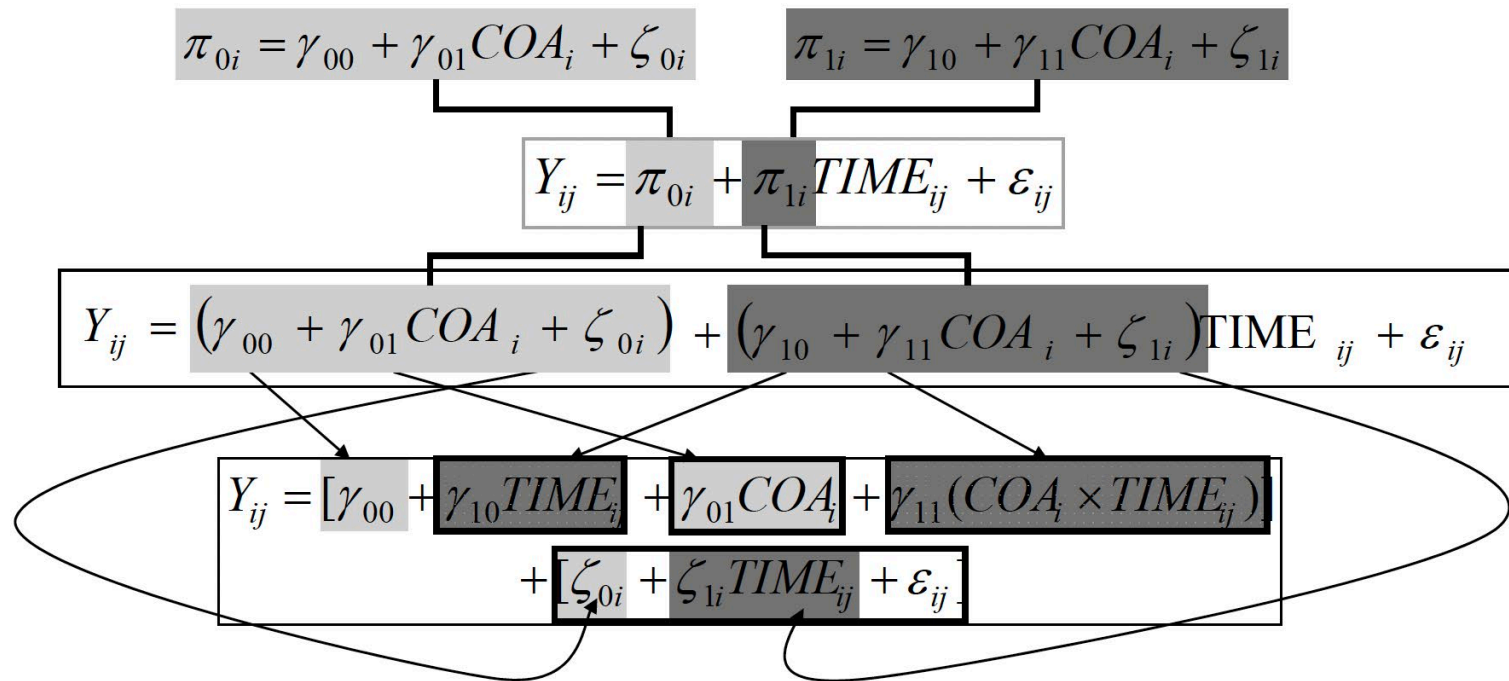
Composite model (alcohol use example)

(Singer and Willet pages 76-81)

- Level – 1
$$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}$$
- Level - 2
$$\pi_{0i} = \gamma_{00} + \gamma_{01}COA_i + \zeta_{0i}$$
$$\pi_{1i} = \gamma_{10} + \gamma_{11}COA_i + \zeta_{1i}$$

- **γ_{00} and γ_{10}** , the level-2 intercepts, represent the population average initial status and rate of change, respectively, for the child of a non-alcoholic (COA = 0).
- If both parameters are 0, the average child whose parents are nonalcoholic uses no alcohol at age 14 and does not change his or her alcohol consumption between ages 14 and 16.
- **γ_{01} and γ_{11}** , the level-2 slopes, represent the effect of COA on the change trajectories (initial status and rates of change), respectively, for children of alcoholics.
- If both parameters are 0, the average child of an alcoholic initially uses no more alcohol than the average child of a non-alcoholic and the rates of change in alcohol use do not differ
- ζ_{0i} and ζ_{1i} , the level-2 residuals, represent deviations of the individual change trajectories around their respective group average trends.

Developing the composite specification of the multilevel model for change by substituting the level-2 submodels into the level-1 individual growth model



The composite specification shows how the outcome depends simultaneously on:

- *the level-1 predictor $TIME$ and the level-2 predictor COA as well as*
- *the cross-level interaction, $COA * TIME$. This tells us that the effect of one predictor ($TIME$) differs by the levels of another predictor (COA)*

The composite specification also:

- *Demonstrates the complexity of the composite residual—this is not regular OLS regression*
- *Is the specification used by most software packages for multilevel modeling*
- *Is the specification that maps most easily onto the person-period data set... $\Rightarrow \Rightarrow \Rightarrow$*

Structural part of level-2 model

(Singer and Willet pages 81-82)

$$\begin{aligned} Y_{ij} &= \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij} \\ &= (\gamma_{00} + \gamma_{01}COA_i + \zeta_{0i}) + (\gamma_{10} + \gamma_{11}COA_i + \zeta_{1i})TIME_{ij} + \varepsilon_{ij}. \end{aligned}$$

Composite multilevel model for change

$$\begin{aligned} Y_{ij} &= [\gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{01}COA_i + \gamma_{11}(COA_i \times TIME_{ij})] \\ &\quad + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}], \end{aligned}$$

- Identifies the cross-level interaction
- More closely aligns with statistical programming
- **The composite and multilevel models are identical mathematically and logically**

Structural part of level-2 model

(Singer and Willet pages 81-82)

Level – 1
$$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}$$

Level – 2
$$\pi_{0i} = \gamma_{00} + \gamma_{01}COA_i + \zeta_{0i}$$
$$\pi_{1i} = \gamma_{10} + \gamma_{11}COA_i + \zeta_{1i}$$

- First part about individual change (within-person change)
- Second part about interindividual change (between-person change)

Structural part of level-2 model

(Singer and Willet pages 81-82)

$$Y_{ij} = [\gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{01}COA_i + \gamma_{11}(COA_i \times TIME_{ij})] \\ + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}],$$

$$\left(\begin{array}{l} \text{Population average} \\ \text{trajectory for the children} \\ \text{of non-alcoholic parents} \end{array} \right) = \gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{01}0 + \gamma_{11}(0 \times TIME_{ij}) \\ = \gamma_{00} + \gamma_{10}TIME_{ij},$$

$$\left(\begin{array}{l} \text{Population average} \\ \text{trajectory for the children} \\ \text{of alcoholic parents} \end{array} \right) = \gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{01}1 + \gamma_{11}(1 \times TIME_{ij}) \\ = (\gamma_{00} + \gamma_{01}) + (\gamma_{10} + \gamma_{11})TIME_{ij},$$

Structural part of level-2 model

(Singer and Willet pages 81-82)

Fixed-effects

$$Y_{ij} = [\gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{01}COA_i + \gamma_{11}(COA_i \times TIME_{ij})] + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}],$$

Random-effects

- More meaningful equation that looks more like the regular regression equation and the STATA syntax
- The measurement of alcohol use depends on the cross-level interaction (level-1 *time* variable and level-2 *COA* variable)
- The composite residuals, or random effects, describe the difference between the observed and the expected value of *Y* for individual *i* on occasion *j*

Stochastic component

(Singer and Willet pages 83-84)

COA = 0

$$\begin{aligned} Y_{ij} &= [\gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{01}0 + \gamma_{11}(0 \times TIME_{ij})] + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}] \\ &= [\gamma_{00} + \gamma_{10}TIME_{ij}] + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}] \\ &= (\gamma_{00} + \zeta_{0i}) + (\gamma_{10} + \zeta_{1i})TIME_{ij} + \varepsilon_{ij}, \end{aligned}$$

The true trajectory of change for individual i

COA = 1

$$\begin{aligned} Y_{ij} &= [\gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{01}1 + \gamma_{11}(1 \times TIME_{ij})] + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}] \\ &= [(\gamma_{00} + \gamma_{01}) + (\gamma_{10} + \gamma_{11})TIME_{ij}] + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}] \\ &= (\gamma_{00} + \gamma_{01} + \zeta_{0i}) + (\gamma_{10} + \gamma_{11} + \zeta_{1i})TIME_{ij} + \varepsilon_{ij}, \end{aligned}$$

The true trajectory of change for individual i

Composite residual: $[\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}]$

- This is the difference between the observed and expected value of $\mathbf{Y}$ for individual i on occasion j

Example: Trajectories of Depression

Data Source: The National Longitudinal Study of Adolescent to Adult Health (Add Health)

Research Question: Are there biological sex differences in trajectories of depression from late adolescence through adulthood?

$$\text{Depression}_{ij} = \pi_{0i} + \pi_{1i}\text{Time}_{ij} + \varepsilon_{ij}$$

Y_{ij} : Mean depression for adolescent i at time j

π_{0i} : Adolescent's i 's true mean depression at wave 1

π_{1i} : The slope (annual rate of change for adolescent i)

ε_{ij} : Random measurement error for adolescent i at time j

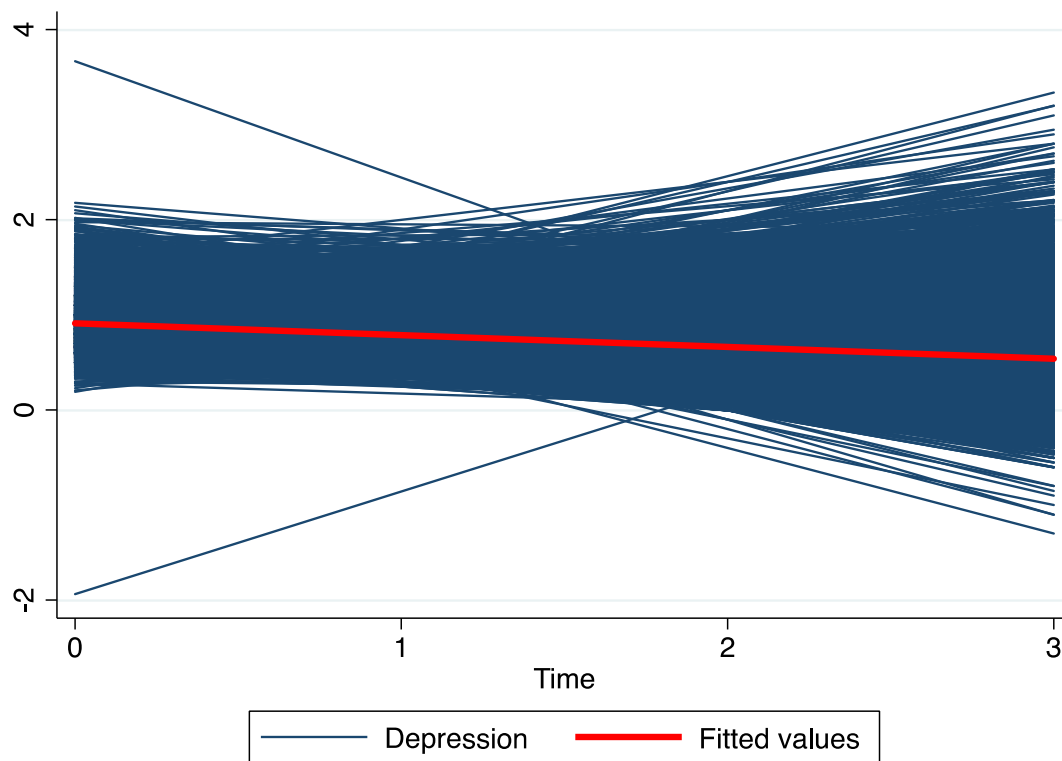
Variable	Obs	Mean	Std. Dev.	Min	Max
dep1	6,472	.8965853	.2738132	.2	2.4
dep2	4,812	.8967997	.2695411	.2	2.4
dep3	4,848	.4971351	.4506727	0	2.777778
dep4	5,104	.6080917	.4691702	0	3

Estimated Intercept and Slope

- N = ~6500
- What do you notice about the slope?
- What do you notice about the intercept?

	int.	slope
intercept	1.0000	
slope	-0.4235	1.0000

Young adults who have high mean depression scores at wave 1 decrease in their level of depression less rapidly over time compared to those start with a low mean depression score

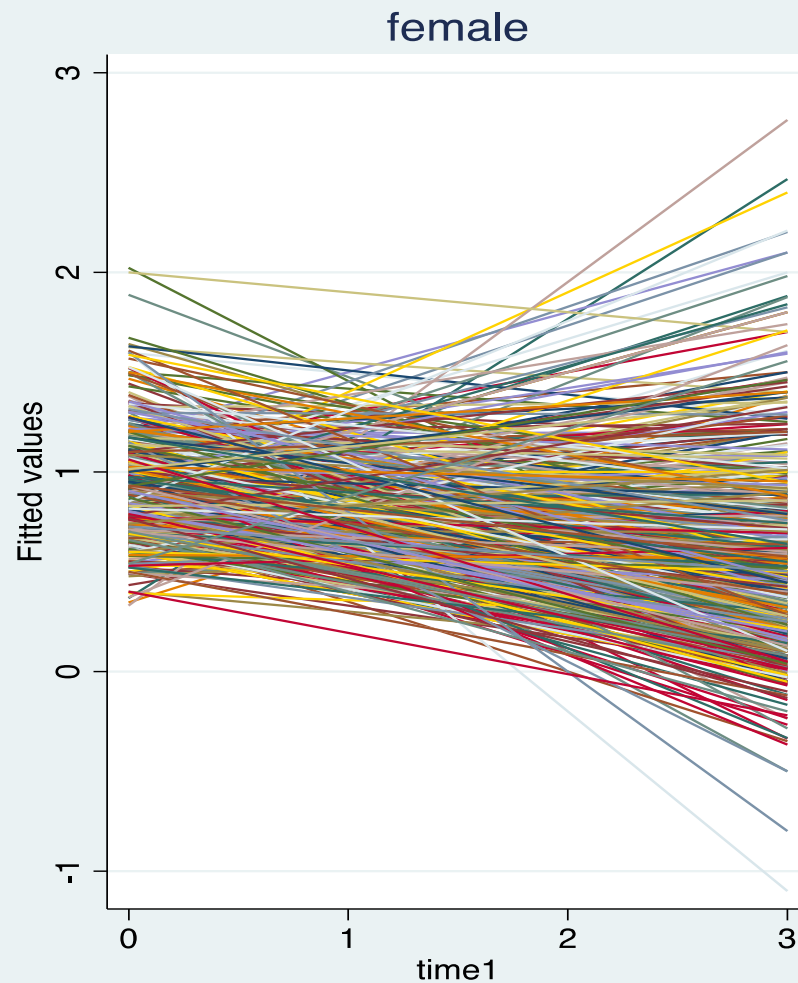
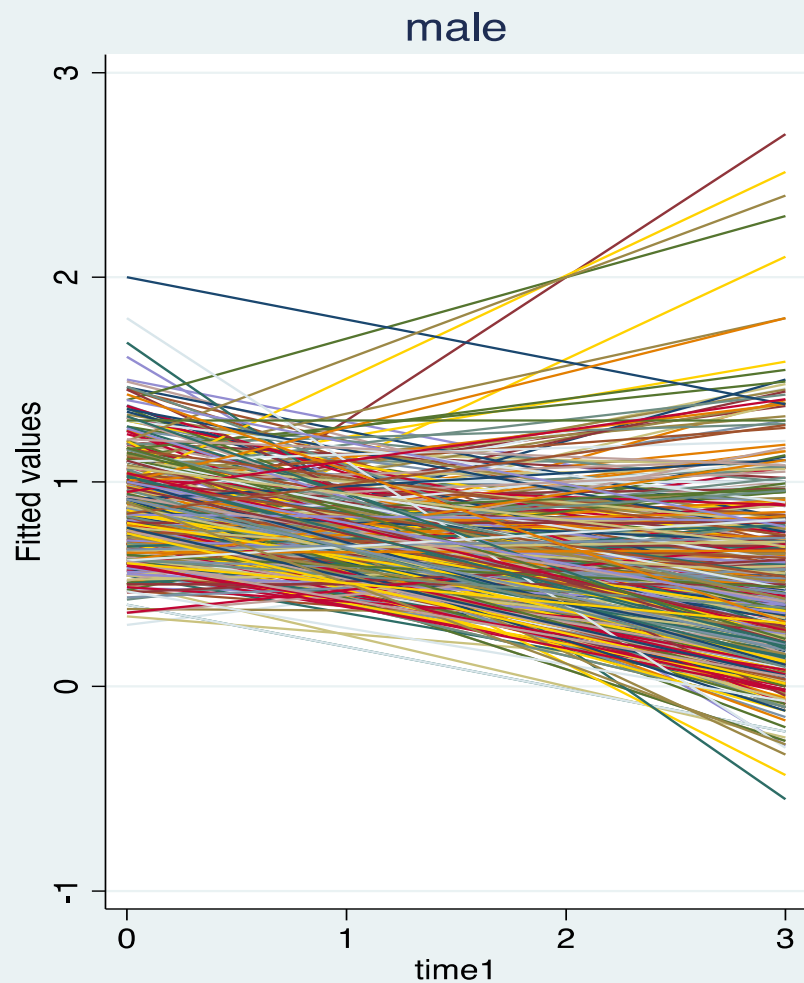


Variable	Obs	Mean	Std. Dev.	Min	Max
intercept	6,169	.9102912	.2709002	-1.933333	3.666667
slope	6,169	-.120331	.1832473	-1.222222	1.077778

Average Estimated Intercept

Average Estimated Slope

Estimated Intercept and Slope



Interpretation of Estimates

Mixed-effects ML regression
Group variable: **id**

Number of obs = **21,236**
Number of groups = **6,497**
Obs per group:
min = **1**
avg = **3.3**
max = **4**
Wald chi2(3) = **2986.80**
Prob > chi2 = **0.0000**

Log likelihood = **-8322.9399**

dep	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
sex	.0781427	.0071871	10.87	0.000	.0640562	.0922291
time1	-.1305959	.0075684	-17.26	0.000	-.1454296	-.1157622
sex_time1	.0039955	.0047006	0.85	0.395	-.0052176	.0132085
_cons	.7927247	.0114936	68.97	0.000	.7701977	.8152518

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
id: Independent				
var(time1)	.012478	.0004645	.0116001	.0134224
var(_cons)	.0143227	.0011764	.0121931	.0168243
var(Residual)	.0902483	.0011347	.0880514	.0924999

Next Class...

- **Singer and Willet Chapter 4**
 - Step 1: Fitting the unconditional models
 - Unconditional mean model
 - Unconditional growth model
 - Step 2: Model building to the “final model”
 - Predictor only model
 - Predictor with control variable model
 - “Final” model

Administrative Stuff

Assignment #2

- Assignment, dataset, and rubric posted to NYUClasses

Class Exercises

Class Exercise #1

1. Do children of alcoholics start drinking alcohol at an earlier age than children of non-alcoholics?
2. Do children of alcoholics increase their alcohol use at a faster rate than children of non-alcoholics?
 - N= 82 adolescents
 - The outcome is alcohol use (**ALCUSE**) in the past year on an 8 point scale with higher numbers indicating more use.
 - The time invariant predictor is dichotomous – child of an alcoholic (**COA**) or not
 - **Age** is the time variable (13-17 years of age)

Class Exercise (cont.) #1

1. Write the level-1 and level-2 equations
 - a. Define the components of the equation (structural and stochastic)
2. Estimate the fixed effects
 - a. Interpret the fixed effects
3. Plot the trajectories

Class Exercise #2

1. Do children who have a high proportion of friends who use alcohol start drinking alcohol at an earlier age than children with a lower proportion of friends who use alcohol?
2. Do children who have a high proportion of friends who use alcohol increase their alcohol use at a faster rate than children with a lower proportion of friends who use alcohol?
 - N= 82 adolescents
 - The outcome is alcohol use (**ALCUSE**) in the past year on an 8 point scale with higher numbers indicating more use.
 - The time invariant continuous predictor is proportion of peers who use alcohol at baseline (**PEER**). Create a mean cutoff for this variable (see UCLA page).
 - **Age** is the time variable (13-17 years of age)

Class Exercise #2

1. Write the level-1 and level-2 equations
 - a. Define the components of the equation (structural and stochastic)
2. Estimate the fixed effects
 - a. Interpret the fixed effects
3. Plot the trajectories